Mini Project

- Implement Cell Transmission Model (CTM)

Individual Work

1. Design your scenario
   - network, demand, incident
   - follow requirements
2. Make CTM simulator
3. Analyze result
4. Share your work
   - Presentation : April 15 (Wed)
   - max 10 min long

Submission

1. Web-based simulator
   - submit the web file
   - you should show it on presentation day
   - if result is not reproduced, minus 10 pts of 100
2. Additional materials (optional)
   - if you needs external files, submit it

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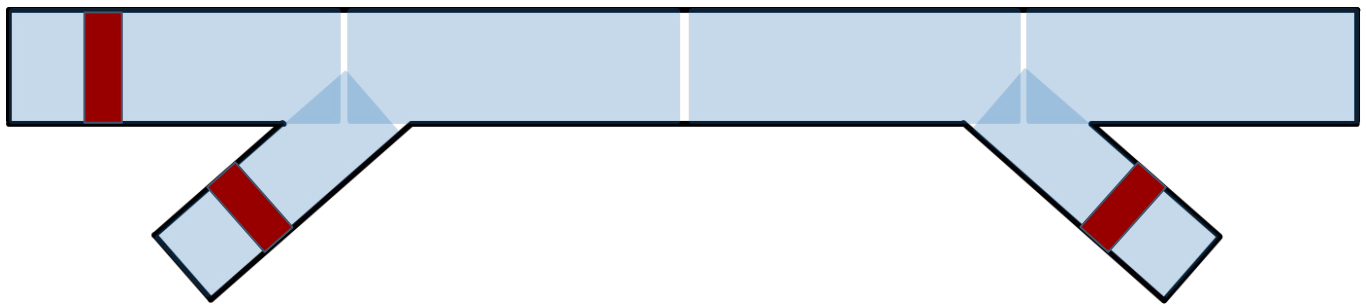

- Network Requirements

  - Highway parameters (general)

    - capacity: 1800~2100 veh/hr/lane
    - free-flow speed: 80~120 km/h
    - wave speed: 10~15 km/h
    - jam density: 150~200 veh/km/lane

  - Include On-Ramp and Off-Ramp

  - No Length Limit for each road (main road, ramps)

  - Minimum # of Cell for main road = 10

  - Simulation Timestep = 5 min

  - Define Position of Flow Detectors (Upstream of Main road, on-/off-ramp)

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- Demand Requirements

  - Based on the flow detector position, define flow through each detector.

  - Comparative analysis for two demands (peak vs off-peak)
  - **Traffic congestion** must occur during peak time

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- Incident Scenario Requirements

  - Simulate incident scenarios for each demand
  - Incident details
    - Assume that lane blockage occurs due to an incident
    - Set the location, time, duration and severity of the incident
      (However, the settings must be maintained identically for all incident scenarios)

  - Compare and analyze the simulation results for each of the scenarios below
    - Off-peak w.o. incident
    - Peak w.o. incident
    - Off-peak with incident
    - Peak with incident

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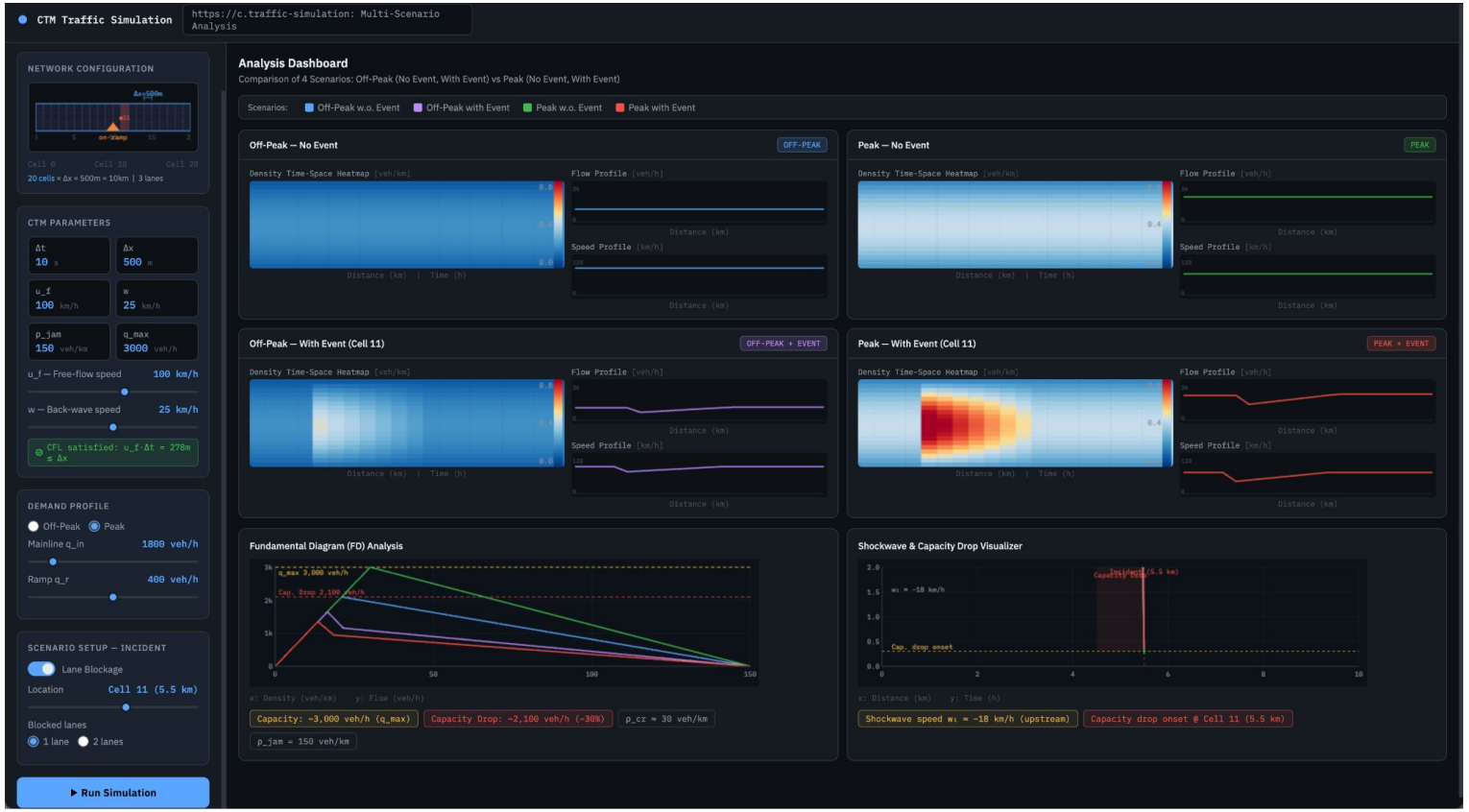
- Presentation Requirements
  - Show and explain about your web-based CTM simulator
  - Check details (following slides)

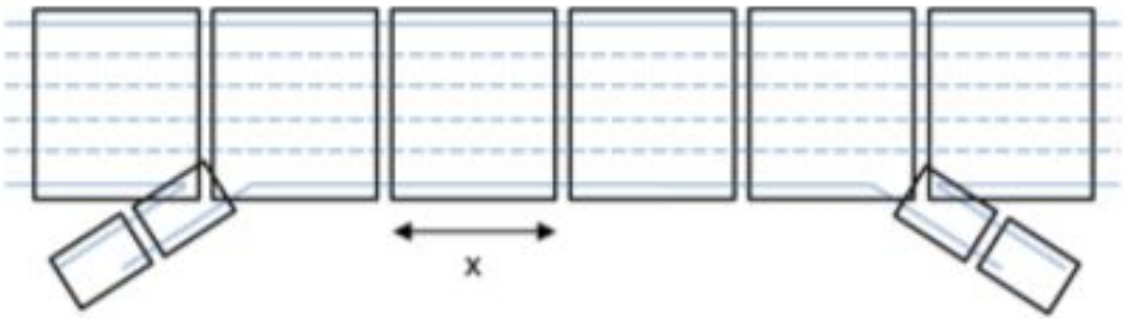
  - Expected Result
  - It's just an example image created by AI
  - so don't follow it because it's not the correct answer



Mini Project

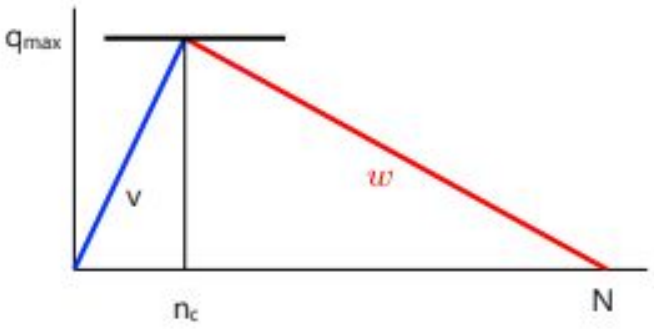
1) Introduce your network and demand to analyze CTM

1-1) Introduce your own network geometry



1-2) Explain and plot the fundamental diagram of the network

Maximum capacity (qmax)	?
Free-flow speed (v)	?
Jam density (N)	?
Wave speed (w)	?



1-3) Explain demand scenarios (how you set simulator for peak, off-peak demand)

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2) Introduce your CTM algorithm

- Follow our AI-policy (check syllabus)
- If you use AI tool for work, then explain details

3) Analyze the result

3-1) Compare and analyze about 4 cases

- Off-Peak w.o. event, Peak w.o. event, Off-Peak with event, Peak with event

3-2) Visualize analyzed results

- Select proper plots to show result like below
  - Basic traffic features (e.g. density, flow, speed)
  - Explain congested situation (e.g. shockwave)